

DMCA — Dynamic Model-Context Alignment

Thesis outline — a Digital-Twin-driven framework for constraint-aware selection and re-alignment of edge AI models

Author: Kevin Omede · M.Sc. candidate, Politecnico di Torino × Beihang University
Supervisor: Prof. Alessandro Simeone (Polito) · **Defence:** September 2026
Contact: kevinomede2002@gmail.com

ABSTRACT

DMCA reframes ML model selection on the industrial edge as a **closed-loop alignment problem** governed by the Digital Twin of the asset. Rather than choosing a model once at design time, the framework continuously re-evaluates the fit between the deployed model and the evolving operating context — hardware, latency SLA, sensor drift, process change — as encoded in an Asset Administration Shell (AAS). The thesis introduces a five-stage pipeline, presents an empirical benchmark of eight time-series foundation models on industrial datasets, and demonstrates a full drift-detection / re-alignment cycle. Implementation in Phase 1 uses a JSON-serialised AAS Type 1 as DT substrate; the architecture is designed to be conformant with IEC 63278 (CDV) and ISO 23247.

THESIS STRUCTURE

Chapter	Title	Content
I	Introduction	Motivation, research questions, contributions, structure of the manuscript.
II	Background & Related Work	Digital Twin and AAS reference models (IEC 63278, ISO 23247, ISO/IEC 30173); time-series foundation models for industrial monitoring; concept-drift detection.
III	DMCA Framework	Five-stage closed loop: (A) DT Profiling via AAS; (B) Multi-criteria model selection (TOPSIS); (C) Edge deployment with four-check quality gate; (D) Two-layer drift detection (ADWIN + decision tree, 7 drift types); (E) Operator-in-the-loop re-alignment.
IV	Experimental Setup	Datasets (CMAPSS FD001, ETT-h1); model registry (8 foundation models); compute environment (NVIDIA T4 on serverless infrastructure, total cost \$0.30).
V	Results	Benchmark across 8 models with 348× latency variation; full DMCA drift-and-recovery cycle (ADWIN detection at t=1380, MAE recovery 3.12 → 0.17 within 20 timesteps); ablation on TOPSIS criteria.
VI	Discussion & Limitations	Static-AAS limitation of Phase 1; single-asset scope; reliance on synthetic drift profiles; pathways to industrial validation.
VII	Conclusion & Future Work	Phase 2 directions: distributed AAS architecture, dynamic submodels as DT service, integration with Eclipse BaSyx, industrial use cases.

SELECTED PRELIMINARY RESULTS

- **Benchmark.** 8 open-source time-series foundation models on CMAPSS FD001 and ETT-h1, on serverless NVIDIA T4 GPU; total compute cost \$0.30. Observed **348× latency variation** across architectures; the most accurate model (TimesFM, MAE = 0.006) is operationally unusable under a 500 ms SLA (measured latency 2039 ms).
- **Drift detection & recovery.** Two-layer detector (ADWIN, $\delta=0.002$) with a v3.1.1 decision tree over 8 hand-crafted features classifies 7 drift types. Held-out validation: **92% accuracy, FAR = 0.000** on 5 drift types × 5 seeds.
- **Closed cycle.** Canonical run: detection at t=1380; chronos-t5-tiny selected; MAE recovers from 3.12 to 0.17 within 20 timesteps.

WHERE PHASE 1 ENDS — OPPORTUNITIES FOR JOINT EXTENSION

Phase 1 was implemented on a **JSON-serialised AAS Type 1** describing a single asset. This is a deliberate simplification to validate the orchestration logic, but it leaves three natural extension directions, all of which align closely with the suggestion kindly outlined by Dr. Belfadel in May 2026:

- Move from a single static AAS to a **distributed AAS** deployed across cloud / fog / edge nodes, with submodels dynamically populated by runtime node status (latency, load, memory, drift indicators).

- Expose the multi-criteria selection step (currently TOPSIS over a static registry) as a **DT service** consuming those submodels and emitting deployment decisions per node.
- Validate the extension on a **real industrial use case** (e.g. through IRT SystemX partners), under realistic SLA and heterogeneity conditions.

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